

# XIAO JIN

Senior Data Scientist · Applied Scientist · Pricing, Revenue Management & Optimization

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## SUMMARY

Ph.D. operations researcher and data scientist with six years building decision systems that reach production. I lead projects end to end — problem framing with commercial stakeholders, modelling, engineering, deployment and post-launch measurement — across airline pricing and revenue management, healthcare operations and logistics. Deep-learning choice models, simulation–optimization, LLM-agent systems and full-stack delivery (Python / PyTorch / Django / React). Currently lead researcher for a corporate-lab portfolio with Singapore Airlines and Scoot, mentoring team members and owning the research agenda across workstreams.

## IMPACT HIGHLIGHTS

- **~8% lift in checked-baggage revenue ( $p < 0.01$ )** — measured in a **randomized field deployment** against the incumbent pricing rule; designed and led the algorithm, now **running in production at Scoot**. Posted prices fell on average while revenue rose. First author on the resulting paper, under review at *Operations Research*.
- **Call abandonment cut from 6% to 4%** — simulation–optimization rostering model **in live monthly use** at the NUHS JMC call centre.
- **~18–21% projected cost saving** on SKU repositioning — simulation and data-driven forecasting backbone for Kuehne+Nagel inventory operations.
- **Transformer-based deep choice model in production** — trained on customer browsing and transactional data to predict ticket purchase; embedded in the SIA Booking Simulator and used by SIA practitioners to stress-test pricing and RM decisions before deployment.

## PROFESSIONAL EXPERIENCE

**SIA–NUS Digital Aviation Corporate Laboratory** — Senior Research Fellow

2024.11 – present

Research Fellow, same lab, 2022.11 – 2024.11 · Singapore

- **Lead researcher, Phase 1 portfolio** (Revenue Management & Network Planning) — own the research agenda and delivery across workstreams; coordinate milestones, deliverables and stakeholder reporting; onboard and technically train new team members.
- **Checked-baggage pricing, Scoot (project lead, shipped)**. Ran randomized price experiments across **1.41M booking sessions** and five baggage tiers, established through reduced-form analysis that substitution is sharply heterogeneous, and built a rational-inattention demand model that prices the whole tier ladder from a handful of interpretable parameters — calibrated by LP so it scales to thousands of sparse segments. **Randomized field deployment: ~8% baggage revenue lift,  $p < 0.01$** , with average posted prices falling. In production.
- **SIA Revenue Prediction Platform (lead researcher, adopted)**. End-to-end simulation platform forecasting revenue over 1-week to 1-year selling horizons under competitor pricing, RM policy and network-level capacity competition. Trained and deployed a **Transformer-based deep choice model** on browsing and transaction data; built the full stack (**Django + React, MongoDB, Celery**) with AI-assisted development. Used by SIA practitioners to validate pricing and RM policy pre-deployment.
- **Call-centre rostering, NUHS JMC (project lead, in live use)**. Discrete-event simulation plus optimization for agent rostering; **abandonment 6% → 4%**; now drives monthly staff planning.
- **Appointment scheduling, National Heart Centre Singapore (project lead)**. Scheduling optimization under demand uncertainty to hold the centre's lead-day service promise.

**Centre for Next Generation Logistics (C4NGL / C4NGP), NUS** — Research Fellow

2020.10 – 2022.11

- **Tuas Port Finger 1 capacity study (project lead)**. Full-scale discrete-event simulation — realistic ship arrivals, container profile distributions, yard layout, crane and AGV dynamics — to estimate annual TEU throughput under candidate layouts. Findings endorsed by the project stakeholder and used to inform layout decisions.
- **Kuehne+Nagel SKU repositioning (core researcher)**. Simulation and simulation-based optimization backbone with data-driven demand forecasting for multi-warehouse inventory repositioning; **~18–21% projected cost reduction**.

**C4NGP, NUS** — Research Engineer

2019.8 – 2020.10

- Built the simulation-analytics and contextual-optimization methods that underpin the doctoral thesis and the later industrial deployments — budget-allocated experimentation under noisy, context-dependent outcomes.

**DB Schenker**, Asia Pacific Regional Office — Intern, Innovation Team

2017.8 – 2017.11

- Scoped and proposed the scenario-based simulation-optimization framework for inventory modelling that became the core of the doctoral thesis.

## TECHNICAL SKILLS

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**Languages** — Python (primary), SQL, C#, C, MATLAB, LaTeX

**ML / AI** — PyTorch, Transformers, deep discrete-choice models, scikit-learn, XGBoost, pandas, NumPy; forecasting; multi-agent LLM systems for optimization modelling; design and causal analysis of randomized pricing experiments

**Optimization & simulation** — Gurobi, CPLEX; stochastic and simulation-based optimization; discrete-event simulation (custom C# / C++ engines, SimPy); ranking & selection

**Engineering** — Django, React, MongoDB, Celery, Git, Docker, AWS / GCP; AI-assisted development with Claude Code for multi-module full-stack delivery

**Domains** — airline revenue management, ancillary and dynamic pricing, assortment optimization, service operations, port and warehouse logistics

## EDUCATION

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**National University of Singapore** — Ph.D., Industrial Systems Engineering & Management

2015.8 – 2020.5

Simulation optimization for context-dependent decisions under noisy, expensive evaluation. GPA 4.21 / 5.00

**Nanjing University** — B.A., School of Management and Engineering

2011.9 – 2015.6

GPA 85 / 100

## SELECTED PUBLICATIONS AND TALKS

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- **First author**, "Pricing Checked Baggage under Rational Inattention: Evidence, Estimation, and Field Deployment." *Under review, Operations Research* — the method deployed at Scoot.
- Jin, X., Shen, Y., & Lee, L. H. "Contextual Optimizer through Neighborhood Estimation for Prescriptive Analytics." *INFORMS Journal on Computing*, 2026.
- Co-author, "Large-Scale Optimization Model Auto-Formulation: Harnessing LLM Flexibility via Structured Workflow" (arXiv:2601.09635, 2026) — **multi-agent LLM system** that generates optimization formulations from natural-language problem descriptions and data, evaluated on a Singapore Airlines revenue-management application.
- "Robust Ancillary Pricing for Airlines: When a Single Mark-up Outperforms Personalisation." INFORMS Annual Meeting, 2025. Invited speaker at **Sabre** on simulation methodology for RM practitioners.
- Ten peer-reviewed conference papers on simulation optimization and budget allocation (Winter Simulation Conference; IEEE CASE, 2017–2022). Full list: [scholar.google.com/citations?user=NffLyjoAAAAJ](https://scholar.google.com/citations?user=NffLyjoAAAAJ)